

Kelly J. Hiersche

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EDUCATION & TRAINING

2026-Current	Massachusetts Institute of Technology , Cambridge, MA Postdoctoral Associate, McGovern Institute of Brain Research *Supported by MIT HEALS Biswas Postdoctoral Fellowship <i>Advisor:</i> Evelina Fedorenko, Ph.D.
2021-2026	The Ohio State University , Columbus OH PhD Candidate, Psychology: Cognitive Neuroscience (Expected graduation Su. 2026) M.S. Cognitive Neuroscience Psychology <i>Thesis:</i> Functional Organization and Modularity of the Superior Temporal Lobe in Children <i>Advisor:</i> Zeynep M. Saygin, Ph.D.
2013-2017	Adelphi University , Garden City, NY Bachelor of Science in Physics Magna Cum Laude Honors College Distinction

PROFESSIONAL MEMBERSHIPS (past and current)

Fetal, Infant, Toddler Neuroimaging; FLUX; Organization Human Brain Mapping; Visions Sciences Society

FELLOWSHIPS, AWARDS, & GRANTS

March 2026	Edward J. Ray Travel Award for Scholarship & Service (\$1,000)
Sept 2023 - 2026	NSF Graduate Research Fellowship (\$37,000 per year) <i>Project:</i> Understanding the selectivity, development, and plasticity of the VWFA
Dec 2025	Center for Cognitive and Behavioral Brain Imaging Research Day <i>Best Poster Presentation:</i> Connectivity-based predictions of putative language regions in the fetal and infant brain
April 2025	Herbert A. Toops Research Award: <i>Most outstanding and creative article published by a Psychology graduate student in 2024:</i> Functional dissociation of the language network and other cognition in early childhood (\$4,000)
April 2025	FoVea Travel and Networking Award (\$1,000)
Dec 2024	Center for Cognitive and Behavioral Brain Imaging Research Day (\$500) <i>Best Oral Presentation:</i> Examining Developmental Variability in Functional Selectivity and Spatial Location of High-Level Visual Regions
March 2024	Edward J. Ray Travel Award for Scholarship & Service (\$1,000)
Nov 2023	Ohio State Psychology Graduate Student Research Forum (\$500) <i>First place presentation;</i> Modularity of Language and Theory of Mind in the Superior Temporal Lobe
April 2023	H. Dean and Susan Regis Gibson Research Award (\$3,000) <i>Project:</i> Understanding the selectivity, development, and plasticity of the VWFA
Feb 2023	NEI Early Career Scientist Travel Grant , Vision Sciences Society (\$500)

Aug 2021	University Fellowship , The Ohio State University (\$30,000)
June 2019	AmeriCorps Education Awards , TFA Placement: Middlesboro, KY (\$5,000)
June 2018	AmeriCorps Education Award , TFA Placement: Middlesboro, KY (\$5,000)
June 2018	NSF Summer Research Fellowship , University of Cincinnati (\$6,500)
June 2016	NSF Summer Research Fellowship , University of Missouri (\$4,500)
May 2017	Ricky Guy Outstanding Physics Senior Award , Adelphi University (\$300)
May 2017	Sigma Pi Sigma , Physics Honor Society, Adelphi University

RELEVANT EXPERIENCE

2021-Present	Graduate Research Assistant , The Ohio State University Supervisor: Zeynep Saygin, Ph.D.
2019-2021	Research Assistant , The Ohio State University Supervisor: Scott Hayes, Ph.D.
2017-2019	Secondary Mathematics Teacher : Middlesboro High School; Teach for America
Summer 2018	NSF Summer Research Fellow , University of Cincinnati Supervisor: John Franco, Ph.D.
2016-2017	Undergraduate Research Assistant , Adelphi University. Supervisor: Dominic Fareri, Ph.D.
Summer 2016	NSF Summer Research Fellow , University of Missouri Supervisor: David Schulz, Ph.D.

PUBLICATIONS AND MANUSCRIPTS

1. DiRubio, L., Li, J., **Hiersche, K.J.**, Aryeetey, N. A., Quatralle, A., Caccese, J., & Saygin, Z. (2026). Structure and function of word laterality in early childhood. *Behavioral Neuroscience*.
2. **Hiersche, K.J.**, Osher, D.E. & Saygin, Z.M. (2026) Functional dissociation of language and theory of mind in the developing superior temporal lobe. *Commun Biol* 9, 558 (2026).
3. Stark, J. H., Lee, A. J., **Hiersche, K. J.**, & Hayes, S. M. (2026). Exploring the relative contributions of multiple modifiable physical health variables to executive function and episodic memory performance in older adults. *Neuropsychology*.
4. **Hiersche, K. J.**, Saygin, Z. M., & Osher, D. E. (2026). Connectivity and function are coupled across cognitive domains throughout the brain. *Network Neuroscience*, 10(1), 80-92.
5. **Hiersche, K. J.**, & Saygin, Z. M. (2026). Leveraging ventral temporal cortex's primary role in object recognition. *Cognitive Neuroscience*, 17(2), 88-91.
6. Olson, H. A., Day, T. K., **Hiersche, K. J.**, Manning, B. L., Bradley, H., Choi, S., ... & Vaughn, K. A. (2025). Utilizing functional neuroimaging to study early language development. *Developmental Cognitive Neuroscience*, 101641.
7. Li, J., **Hiersche, K. J.**, & Saygin, Z. M. (2024). Demystifying visual word form area visual and nonvisual response properties with precision fMRI. *iScience*, 27(12).
8. Cloud, J., Stark, J., Hasselbach, A. N., **Hiersche, K. J.**, Salat, D. H., Juttukonda, M. R., & Hayes, S. M. (2024). Exercise-induced reductions in central command network cerebral blood flow assessed with arterial spin labeling MRI. *Imaging Neuroscience*, 2, 1-13.
9. **Hiersche, K.**, Schettini, E., Li, J., & Saygin, Z. (2024). Functional dissociation of the language network and other cognition in early childhood. *Human Brain Mapping*, 45(9), e26757.

10. Schettini, E., **Hiersche, K. J.**, & Saygin, Z. M. (2023). Individual variability in performance reflects selectivity of the multiple demand network among children and adults. *Journal of Neuroscience*, 43(11), 1940-1951.
11. Stauder, M., **Hiersche, K. J.**, & Hayes, S. M. (2023). Examining cross-sectional and longitudinal relationships between multidomain physical fitness metrics, education, and cognition in Black older adults. *Aging, Neuropsychology, and Cognition*, 1-15.
12. Stark, J., **Hiersche, K. J.**, Yu, J. C., Hasselbach, A. N., Abdi, H., & Hayes, S. M. (2022). Partial Least Squares Regression analysis of Alzheimer's disease biomarkers, modifiable health variables, and cognitive change in older adults with Mild Cognitive Impairment. *Journal of Alzheimer's Disease*, 1-19.
13. Stark, J., Palombo, D. J., Hayes, J. P., **Hiersche, K. J.**, Hasselbach, A. N., Hayes, S. M., & Alzheimer's Disease Neuroimaging Initiative. (2021). Partial least squares analysis of Alzheimer's disease biomarkers, modifiable health variables, and cognition in older adults with mild cognitive impairment. *Journal of the International Neuropsychological Society*, 28(8), 781-789.

PREPRINTS & IN PREPARATION

1. Li, J., **Hiersche, K.J.**, Aryeetey, N., Quatrala, A., Resnick, P., Saygin, Z.M. Literacy acquisition and functional connectivity drive the emergence and continued development of the VWFA. (submitted).
2. Melville, M., Stark, J., **Hiersche, K.J.**, Esterman, M., Hayes, S.M. (in revision). Examining the Impact of Light and Vigorous Acute Aerobic Exercise on Pattern Separation and Sustained Attention in Young Adults. OSFF Preprints, (2024-08).
3. Rydel, L., **Hiersche, K.J.**, Saygin, Z.M. (in preparation). A Multivariate Analysis of Visual and Auditory Language Responses in the Proto-VWFA of Pre-Reading Children.

TALK PRESENTATIONS

1. **Hiersche, K.J.**, Rydel L.A., Saygin, Z.M. (May 2026). *A multivariate approach to understanding the emergence of the visual word form area*. Vision Sciences Society, St. Pete Beach, FL.
2. **Hiersche, K.J.** (Oct 2025). *The Developing Language Network*. EvLab, MIT, Boston, MA.
3. **Hiersche, K.J.**, Wang, X., Quatrala, A., Dickerson, J., Holt, R.F., Saygin, Z.M. (Sept 2025). *Predicting Activation in the Fetal and Infant Brain*. CCBS Annual Retreat, Columbus, OH.
4. **Hiersche, K.J.** & Saygin, Z.M. (Sept 2025). *Perinatal neurobiological markers of cognitive outcome in normotensive and preeclampsia pregnancies*. CTSI Research Showcase, Columbus, OH.
5. **Hiersche, K.J.** (Feb 2025). *Examining the association between connectivity & function*. Cognitive Neuroscience Brownbag, Ohio State University, Columbus, OH.
6. **Hiersche, K.J.**, Quatrala, A., Saygin, Z.M. (Dec 2024). *Examining developmental variability in functional selectivity and spatial location of high-level visual regions*. CCBBI Research Day, Columbus, OH.
7. **Hiersche, K.J.** (March 2024). *Language and other cognition in the developing brain*. Social Cognitive Neuroscience Lab Meeting, MIT, Boston, MA.
8. **Hiersche, K.J.**, (March 2024). *Neural Development of Language in Infants and Toddlers*. Cognitive Neuroscience Brownbag, Ohio State University, Columbus, OH.
9. **Hiersche, K.J.**, (Sept 2023). *Language and Theory of Mind in the Developing Superior Temporal Lobe*. Undergraduate Cognitive Neuroscience Club Meeting, Columbus, OH.
10. **Hiersche, K.J.**, (Sept 2023). *Modularity of Language and Theory of Mind in the Superior Temporal Lobe*. CCBS Annual Retreat, Columbus, OH.

11. **Hiersche, K.J.**, (March 2023) *Examining the functional organization and modularity of the Superior Temporal Lobe*. Cognitive Neuroscience Brownbag, Ohio State University, Columbus, OH.
12. **Hiersche, K.J.**, (Feb 2022). *The Selectivity of the Language Network in Children*. Cognitive Neuroscience Brownbag, Ohio State University, Columbus, OH.
13. **Hiersche, K.J.** (2019). *Challenge Based Learning Encryption with Algebra II*. Research Experience for Teachers Community of Practice, University of Cincinnati, Cincinnati, OH.

POSTER PRESENTATIONS

1. **Hiersche, K.J.**, Wang, X., Quatrala, A., Dickerson, J., Taylor, G.H., Hamilton, J., Constantine, M., Holt, R.F., Saygin, Z.M., (Sept, 2025). *Connectivity-based prediction of putative language regions in the fetal brain*. Poster Presentation. FLUX, Dublin, Ireland.
2. **Hiersche, K.J.**, Saygin, Z.M., (May, 2025). *Examining Variability of High-Level Visual Categories Across Development*. Poster Presentation. Vision Sciences Society, St. Pete Beach, FL.
3. **Hiersche, K.J.**, Saygin, Z.M. (Sept, 2024). *Examining the functional organization, specificity, and laterality of language and theory of mind (ToM)*. FLUX, Baltimore, MD.
4. Quatrala, A., **Hiersche, K.J.**, Saygin, Z.M. (Sept, 2024). *Unraveling Developmental Variability in Functional Selectivity and Spatial Localization of High-Level Visual Regions of the Ventral Temporal Cortex*. FLUX, Baltimore, MD.
5. **Hiersche K.J.**, Saygin, Z.M. (Sept, 2024). *Asymmetries in structure and connectivity precede functional specialization for language in the infant brain*. Fetal, Infant, and Toddler Neuroimaging, Baltimore, MD.
6. **Hiersche, K.J.**, Saygin, Z.M. (Sept 2024). *Neural Development of Language in Infants and Toddlers*, Center for Cognitive and Brain Sciences Fall Retreat, Columbus, OH.
7. **Hiersche, K.J.**, Saygin, Z.M. (June, 2024). *Neural Development of Language in Infants and Toddlers*, Organization for Human Brain Mapping, Seoul, Korea.
8. **Hiersche, K.J.**, Quatrala, A., Saygin, Z.M. (May 2024). *Examining Variability of High-Level Visual Categories Across Development*, Vision Sciences Society, St. Pete Beach, FL
9. **Hiersche, K.J.**, Saygin, Z.M. (July 2023). *Examining modularity in the superior temporal sulcus in young children*. Organization for Human Brain Mapping, Montreal, CA.
10. **Hiersche, K.J.**, Li, J., Saygin, Z.M. (May 2023). *The functional profile of the ventrotemporal cortex to high-level vision, language, and attention*. Vision Sciences Society, St. Pete Beach, FL.
11. **Hiersche, K.J.** & Saygin, Z.M. (Dec 2022). *The development of language and social cognition in young children*. Center for Cognitive and Behavioral Brain Imaging Research Day, Columbus, OH.
12. **Hiersche, K.J.**, Li, J., Saygin, Z.M. (May 2022). *The selectivity and development of the visual word form area and frontotemporal language network in pre-readers and beginning readers*. Vision Sciences Society, St. Pete Beach, FL.
13. **Hiersche, K.J.**, Li, J., Saygin, Z.M. (March 2022). *The selectivity of the visual word form area and longitudinal development of the frontotemporal language network in pre-readers and beginning readers*. Chronic Brain Injury Research Day, Columbus, OH.
14. **Hiersche, K.J.**, Stark, J., Hasselbach, A., & Hayes, S.M. (Feb 2022). *The Examination of Multiple Modifiable Fitness Variables and Cognition in Healthy Older Adults*. Virtual International Neuropsychological Institute, New Orleans, LA.
15. **Hiersche, K.J.** & Saygin, Z.M. (Dec 2021). *The selectivity and development of the frontotemporal language network in early childhood*. Center for Cognitive and Behavioral Brain Imaging Research Day, Columbus, OH.

16. **Hiersche, K.J.**, Stark, J., Hasselbach, & Hayes, S.M. (March 2021). *Examining the Relationship between Modifiable Health Variables and Cognition in Healthy Older Adults*. Chronic Brain Injury Research Day, Columbus, OH.
17. **Hiersche, K.J.**, Stark, J., Hasselbach, A.N., Kloos, A., Kegelmeyer, D., Esterman, M., & Hayes, S.M. (March 2020). *Examining the Relationship between Physical Health and Cognition in Adults*. Chronic Brain Injury Research Day, Columbus, OH.
18. Hayes, S.M., Williams, V.J., Salat, D., Hasselbach, A.N., **Hiersche, K.J.**, Stark, J. (March 2020). *Examining the feasibility of aerobic exercise during brain functional MRI*. Chronic Brain Injury Research Day, Columbus, OH.
19. **Hiersche, K.J.** (2018). *Cybersecurity and Data Encryption*. 2018 STEM Conference, North Kentucky University, Highland Heights, KY.
20. **Hiersche, K.J.**, Mesewicz, A., Kapoor, S., & Franco, J. (2018). *Cybersecurity and Data Encryption*. Research Experience for Teachers Culmination Event, University of Cincinnati, Cincinnati, OH.
21. **Hiersche, K.J.**, & Mesewicz, A. (2018). *Secure Cyberspace*. Oral Presentation at Research Experience for Teachers Culmination Event, University of Cincinnati, Cincinnati, OH.
22. **Hiersche, K.J.**, Lane, Brian J., Kick, Daniel R., Schulz, David, J. (2017). *Relative Ionic Current Contributions to the Driver Potentials of Large Cells in the H. americanus Cardiac Ganglion*. Adelphi Research Conference, Adelphi University, Garden City, NY.
23. **Hiersche, K.J.**, Lane, Brian J., Kick, Daniel R., Schulz, David, J. (2016). *Relative Current Contributions to the Driver Potentials of Large Cells in the Crustacean Cardiac Ganglion*. NSF REU: Summer Research Symposium, University of Missouri, Columbia, MO.

COMMUNITY OUTREACH

Sp 2025	STEAMM RISING: volunteer at outreach event for Columbus City School teachers
Fall 2024	Presentations on Fetal Brain development: bi-weekly community outreach presentations at Mother to be classes
Fall 2024-Fall 2025	Volunteer Soccer Coach: Westerville Middle School Girls
Summer 2024/25	Assistant Program Manager: Center for Cognitive and Brain Sciences (CCBS): Humanities & Cognitive Sciences Summer Institute (HumCog)
Fall 2022-Summer 2023	Graduate Research Assistant: Center for Cognitive and Brain Sciences
Summer 2022	Organizer & Volunteer: CCBS HumCog Summer High School Program
Spring 2022/23/24	Organizer & Volunteer: COSI Science Festival Lab Booth
Spring 2022/23	Presenter: Metro North 7th Annual Career Fair for High & Middle Schoolers
Spring 2022	Presenter: Nu Rho Psi Graduate Student Panel
Sp /F 2022, Sp 2023/24	Program Manager: Soccer & Literacy Intervention, SproutFive Preschools
Summer 2019, 2021	Volunteer: Ohio Senior Olympics, Westerville, OH

LEADERSHIP & SERVICE

Fall 2023-Summer 2025	President: CCBBI Student Group
Su 2024/25	Assistant Program Manager for CCBS HumCog High School Summer Camp
Fall 2021-Summer 2023	Treasurer: CCBBI Student Group
Fall 2021-Summer 2023	Member: Center for Cognitive and Behavioral Brain Imaging (CCBBI) Talk Series

MENTORSHIP

Fall 2024-Spring 2026	Undergraduate Student Mentor: <i>Lauren Rydel</i>
Fall 2025	Undergraduate Student Mentor: <i>Isabel Beck</i>
Fall 2023-Spring 2026	Post-Bac Mentor: <i>Anna Quatralle</i>
Summer 2023	Neuroscience R25 Bench Mentor, High School Student: <i>Alexander Olegnowicz</i>
Fall 2022-Spring 2024	Undergraduate Student Mentor: <i>Ajuna Mwesigye</i>
Fall 2021-Spring 2022	Undergraduate Student Mentor: <i>Luke Wisniewski & Brenden Dalton</i>

TEACHING

Sept 2025	Guest Lecture: Developmental Cognitive Neuroscience: High Level Vision
Oct 2024	Guest Lecture: Intro to Cognitive Neuroscience: Motor
Oct 2023	Guest Lecture: Intro to Cognitive Neuroscience: Emotion
Oct 2023	Guest Lecture: Intro to Psychology: Language
March 2024	Guest Lecture: Intro to Cognitive Neuroscience: Cognitive Control

ADHOC REVIEWING

Developmental Cognitive Neuroscience, Cerebral Cortex, eLife, Journal of Neuroscience

POSTERS AND PRESENTATIONS: COLLABORATOR

1. Wang, X., **Hiersche, K.J.**, Quatralle, A., Sponseller, E., Holt, R.F., Saygin, Z.M. (2024) *Fetal structural and functional fMRI scanning: experiences, challenges, and future development*. CCBBI Research Day.
2. Rydel, L., **Hiersche, K.J.**, Saygin, Z.M. (2024). *Investigating Visual and Auditory Language Responses in the Ventral Temporal Cortex of Pre-Reading Children. A Multivariate Analysis*. CCBBI Research Day.
3. Bradley, L., **Hiersche, K.J.**, Quatralle, A., Saygin, Z.M. (2024). *Interhemispheric Differences in Sulcal Depth Across Development*. CCBBI Research Day.
4. Li, J., **Hiersche, K.J.**, Saygin, Z.M. (2023). *The visual word form area is indeed visual: precision fMRI with 4 tasks to characterize specificity*. Organization for Human Brain Mapping, Montreal, CA.
5. Stark, J., **Hiersche, K.J.**, & Hayes, S.M. (2023). *Exploring the differential importance of modifiable fitness variables on cognitive performance in older adults*. International Neuropsychological Society, San Diego, CA.
6. Melville, M.A., Stark, J., **Hiersche, K.J.**, Esterman, M., & Hayes, S.M. (2023). *Light and vigorous bouts of acute aerobic exercise positively impacts sustained attention and inhibition but not pattern separation in young adults*. International Neuropsychological Society, San Diego, CA.
7. Stauder, M.J., **Hiersche, K.J.** & Hayes, S.M. (2023). *Cognitive reserve and gait speed are associated with cognitive performance in Black/African American older adults*. International Neuropsychological Society, San Diego, CA.
8. Hayes, S.M., Williams, V.J., Salat, D., Hasselbach, A.N., **Hiersche, K.J.**, Stark, J. (2023). *Assessing the feasibility of implementing a cardiorespiratory exercise during functional MRI*. International Neuropsychological Society, San Diego, CA.
9. Stark, J., **Hiersche, K.J.** & Hayes, S.M. (2023). *Exploring the differential importance of modifiable fitness variables on cognitive performance in older adults*. International Neuropsychological Society, San Diego, CA.

10. Dalton, B.K., **Hiersche, K.J.**, Caccese, J., Schettini, E., Saygin, Z.M. (2022). *The effect of repetitive neurotrauma on development*. Center for Cognitive and Behavioral Brain Imaging Research Day, Columbus, OH.
11. Hayes, S.M., Stark, J., **Hiersche, K.J.** & Hayes, J.P. (2022). *Modifiable physical fitness is associated with executive function in Black older adults*. Oral Paper Presentation at National Academy of Neuropsychology, Denver, CO.
12. Schettini, E., **Hiersche, K.J.**, & Saygin, Z.S. (2022). *Comparison of selectivity of Multiple Demand network between children and adults*. Cognitive Neuroscience Society, San Francisco, CA.
13. Stauder, M.J., **Hiersche, K.J.**, & Hayes, S.H. (2022). *Exploring associations between cognitive reserve, physical fitness, and cognition in Black/African American Older Adults*. Chronic Brain Injury Day, Columbus, OH.
14. Huston, C., Shin, A., Hasselbach, A., **Hiersche, K.J.**, & Hayes, S.H. (2022). *Exercise-induced alterations in neural BOLD signal*. Chronic Brain Injury Day, Columbus, OH.
15. Melville, M., **Hiersche, K.J.**, Hasselbach, A.N., Shin, A., & Hayes, S.H. (2022). *The Impact of exercise intensity on executive function performance*. Chronic Brain Injury Day, Columbus, OH.
16. Stark, J., **Hiersche, K.J.**, Hasselbach, A.N., Hayes, J.P., & Hayes, S.H. (2022). *Partial least squares regression analysis of Alzheimer's disease biomarkers, modifiable health variables, and longitudinal cognition in older adults with Mild Cognitive Impairment*. Chronic Brain Injury Day, Columbus, OH.
17. Schettini, E., **Hiersche, K.J.**, & Saygin, Z.S. (2021). *Comparison of selectivity of Multiple Demand network between children and adults*. Center for Cognitive and Behavioral Brain Imaging Research Day, Columbus, OH.
18. Stark, J., **Hiersche, K.J.**, Hasselbach, A.N., Hayes, J.P., & Hayes, S.H. (2022). *Partial least squares regression analysis of Alzheimer's disease biomarkers, modifiable health variables, and longitudinal cognition in older adults with Mild Cognitive Impairment*. Virtual International Neuropsychological Society Conference.
19. Sagar, A., **Hiersche, K.J.**, Stark, J., & Hayes, S.M. (2021). *Examining Associations Between Inflammation, Modifiable Health Variables, and Cognition in Black Older Adults*. Denman Undergraduate Research Forum, Columbus, OH.
20. Stark, J., Palombo, D.J., Hayes, J.P., **Hiersche, K.J.**, Hasselbach, A.N., & Hayes, S.H. (2021). *Partial Least Squares analysis of Alzheimer's disease biomarkers, modifiable health variables, and cognition in older adults with Mild Cognitive Impairment*. International Neuropsychological Society Conference.
21. Hasselbach, A. N., **Hiersche, K. J.**, Osher, D., & Hayes, S. M. (2020). *Fitness and Cortical Thickness in Cognitively Healthy Young Adults*. Chronic Brain Injury Research Day, Columbus, OH.
22. Sagar, L., Tacsar, N., Berry, A.M., **Hiersche, K.J.**, Stark, J., Hasselbach, A.N., & Hayes, S.M. (2020). *Exploring Interactions Between Physical Activity, Fitness, Sleep, and Cognitive Performance in Young and Older Adults*. Chronic Brain Injury Research Day, Columbus, OH.
23. Stark, J., **Hiersche, K.J.**, Hasselbach, A.N., Palombo, D.J., & Hayes, S.M. (2020). *Modifiable Physical/Health Variables and Cognition in ADNI1: a cross-sectional analysis using Partial Least Squares*. Chronic Brain Injury Research Day, Columbus, OH.